

Shada — Make and Print List

4 pieces to print, 5 to make by hand. Everything on this sheet is built in-house; everything bought is on the shopping list.

THIS FILE IS GENERATED FROM `outfits.yaml`. That file's components: `list` is the source of truth. Do not edit this sheet to record a decision — edit `outfits.yaml` and run `./tools/build-lists shada`. Sizes are approximate and several are estimated from assumed dimensions; confirm against the real patterns and against the performer before committing to a print run.

3D printed

Printed parts come out identical, so **all mismatch is painted on**. That is the rule this whole section turns on.

1. Hex plates — the raw scale

Printed $\times \sim 440$. *Not started.*

Size	10 to 15 mm across the flats — regular equilateral hexagons, ONE printed master repeated ~ 440 times. That is much finer than it sounds: a hexagon 12.5 mm across covers about 1.35 cm ² . IF YOU CAN COUNT THE PLATES AT A GLANCE THEY ARE WRONG. Each needs TWO PUNCHED HOLES for lacing.
Colour	 Dull grey steel — the base tone, and the commonest
Finish	PRINTED IDENTICAL, SO ALL MISMATCH IS PAINTED ON. The five tones above are mismatched alloys oxidising at different rates, not a decorative scheme — mix them irregularly and never in a pattern. Each plate carries a RAISED SERPENT RIDGE, and at 12 mm the relief alone will not carry it: it needs a wash pulled into the ridge and a dry-brush off the top, or it will not read at all on camera.
Method	THREE OR FOUR RIDGE DIES REPEATING across a panel — never a unique curl per plate. PRINT AND FINISH A FULL PATCH BEFORE COMMITTING to a run of several hundred; this is not something to discover a problem in halfway through. DO NOT PRINT PRE-LINKED SHEETS. The pressure to will be enormous and it is exactly the shortcut that kills the flex — all four cloth pieces must move like heavy cloth. If assembly time is the problem, REDUCE THE PATCH AREAS rather than link the plates.

Palette — all five, mixed irregularly.

Colour	Where it goes
 Dull grey steel	The base tone and the commonest across all four cloth pieces.
 Blackened iron	Scattered singly, never in runs. Reads as the oldest metal.
 Dark bronze	THE THIGH PATCH ONLY. The one warm thing in the whole costume and the only place this colour appears — do not let it drift onto the flank panels.
 Pale worn pewter	The highlight tone. Edges and high points that have rubbed.
 Rust-red oxidised iron	Sparingly, and at the edges of a panel where damp sits. NO BRASS AND NO VERDIGRIS anywhere on this costume.

From the specification: 10 to 15 mm across, regular equilateral hexagons, ONE printed master repeated.

Re-confirmed 2026-08-01 against a generation that came back three to four times too coarse. If you can count the plates at a glance they are wrong. Printed identical, so ALL mismatch is painted on — mismatched alloys oxidising

at different rates: dull grey steel, blackened iron, dark bronze, pale worn pewter, rust-red oxidised iron. NO BRASS, NO VERDIGRIS. Each carries a RAISED serpent ridge, three or four dies repeating across a panel, never a unique curl per plate.

2. Gauntlet — her RIGHT forearm, solid printed shell

Printed ×1. Not started.

Size	Covers ~200 cm ² of her right forearm — wrist to below the elbow. Printed as ONE SHELL, which is what removes the ~150 hand-laced plates this area would otherwise have cost.
Colour	 Dull grey steel
Accent	 Dim amber telltales, a cluster at the wrist
Finish	MUST NOT MATCH the other four pieces — it is scavenged from somewhere else and the mismatch is the point. THE RAISED SWIRL IS THE WHOLE JOB HERE: nothing else distinguishes this from a moulded tech panel, so the wash and dry-brush that make the relief read are not optional. The telltales are dim and warm, not bright LEDs.
Method	The one piece in the costume that does not flex, and the only one where a hard printed shell is correct. Print the cuff separately if it fights the build plate; the cuff reads warm brown as bought leather and needs the same taupe dye job as the belts.

From the specification: Dull grey steel, a cluster of dim amber telltales at the wrist. The one piece that does not flex. MUST NOT MATCH the other four pieces. Printing it is worth ~150 hand-laced plates, each of which would have needed two punched holes — the largest single saving in this build, and it survives every revision because it is a property of the piece.

3. WESTAR-35 blaster pistol

Printed ×1. Not started.

Size	~300 to 330 mm overall — the WESTAR-35 is a long-barrelled pistol and the length is most of its silhouette. ONE PRINTED WEAPON CAN SERVE TWO CHARACTERS: see Costume-Build-Method. Check before printing a second.
Colour	 Gunmetal, dulled
Accent	 Worn steel at the muzzle, grip and draw edges
Finish	WELL USED, NEVER FACTORY FRESH — the opposite instruction to Jasu's A180, which is the one clean weapon in the film. Rub the finish back where a hand and a holster mouth have worn it, and let the recesses stay dark.
Method	On her RIGHT side, hip or thigh. She is right handed.

From the specification: On her RIGHT side, hip or thigh. Well used, never factory fresh.

4. Combat knife

Printed ×1. Not started.

Size	~200 to 250 mm overall, blade ~120 to 150 mm. A working tool, not a fighting trophy.
Colour	 Dull worn steel
Accent	 Dark wrapped or scaled grip

Finish	WELL USED. Sharpen-marks along the edge and a rubbed-back flat; leave the spine and the ricasso darker. It must never look ornamental.
Method	On her LEFT hip, off the drawing side. PRINT IT BLUNT — a costume knife carried against the body needs a rounded edge and a rounded point regardless of how it reads on camera.

From the specification: On her LEFT hip. Well used.

Made by hand — not printed

Do not print these. Each one has to move, flex or be fitted to the performer, and a printed shell is the specific failure recorded against them.

1. Flank panels — a laced pair, centre front over the ribs

Made ×2. *Not started.*

Size	~150 cm ² each, ~110 plates each — HALF THE PLATE COUNT IN THE COSTUME. About twelve plates across a panel. PATTERN THESE FIRST: their area depends on a pattern nobody has cut yet, so they are both the biggest number in this build and the least certain one.
Colour	 Mixed from the hex-plate palette — unmatched to each other
Finish	FIVE TO TEN PLATES ABSENT from each, along the edges, waist and hip where a panel catches and flexes. LACE COMPLETE, THEN REMOVE — scattered absence reads as wear, regular absence reads as a pattern. A thin dark seam between neighbours.
Method	Laced onto hand-cut leather-and-fabric backing. TESSELLATED FLAT IN ONE PLANE, NEVER SHINGLED. They sit over her ribs where she breathes, so they must move like heavy cloth. Punched eyelets down both facing edges, a leather thong long enough to lace the full height, and two strap-and-buckle fixings at the back — the lacing is visible and structural and cannot be faked with a hidden fastening. NO BREASTPLATE: chest and sternum stay plain cloth between the panels.

From the specification: ~110 plates each, and half the plate count in the costume. PATTERN THESE FIRST — their area depends on a pattern nobody has cut yet. Unmatched to each other. Laced onto hand-cut leather-and-fabric backing, tessellated flat in one plane, never shingled, a thin dark seam between neighbours. FIVE TO TEN PLATES ABSENT from each, along the edges, waist and hip where a panel catches and flexes. Lace complete, then remove — scattered absence reads as wear, regular absence reads as a pattern. NO BREASTPLATE: chest and sternum stay plain cloth between the panels.

2. Scale cap — her LEFT shoulder

Made ×1. *Not started.*

Size	~120 cm ² , ~90 plates. A PALM'S WIDTH — the point of the shoulder only, and no further down the arm.
Colour	 Mixed from the hex-plate palette
Finish	Three to five plates absent.
Method	MOVES LIKE HEAVY CLOTH, NOT PLATE — it has to hang loose and swing. Pre-linked printed sheets are FORBIDDEN here.

From the specification: ~90 plates. A PALM'S WIDTH — the point of the shoulder only. Three to five plates absent. Moves like heavy cloth, not plate.

3. Scale patch — her LEFT outer thigh

Made ×1. *Not started.*

Size	~180 cm ² , ~130 plates.
Colour	 Dark bronze
Finish	THE ONE WARM THING IN THE WHOLE COSTUME, and the only place this colour appears. Everything else is grey, black, pewter and rust.
Method	Must move like heavy cloth. Pre-linked printed sheets are FORBIDDEN here.

From the specification: ~130 plates. DARK BRONZE — the one warm thing in the whole costume, and the only place that colour appears.

4. Holster and knife sheath — her RIGHT thigh

Made ×2. *Not started.*

Size	BUILD AROUND THE PRINTED WESTAR-35 AND THE PRINTED KNIFE, and not before they exist. The WESTAR's long barrel sets the holster depth and it is longer than a generic pistol pattern allows for.
Colour	 Taupe — grey-brown, cool
Finish	TAUPE IS A DYEING JOB, NOT A PURCHASE. Leather reads warm brown as bought and the costume drifts back toward brown on almost every image; it will do the same on the bench. Dye and knock the warmth out, then check it beside the belts in daylight.
Method	Holster on her RIGHT thigh, on the drawing side. The knife sheath goes on her LEFT hip — the two are made together and worn apart.

From the specification: Taupe leather, on the drawing side. Built around the WESTAR-35.

5. Scale make-up — face, neck, collarbone, shoulders, arms, hands

Made ×1. *Not started.*

Size	FINE — much finer than the armour plates, and a separate discipline from them. Face, neck, collarbone, shoulders, arms and hands.
Colour	 Skin, with the scale pattern read in raking light
Accent	 Reptilian pupil — CONTACT LENS on a human performer
Finish	IT IS SKIN AND IT READS AS SKIN. Never painted on and never a printed pattern. Locked against scale_portrait.png, NOT against the costume reference.
Method	THE PUPIL IS THE SINGLE HIGHEST-VALUE ELEMENT IN HER MAKE-UP. The eye keeps its human shape, lids, lashes and expression, and only the pupil is wrong. NOT A BOUGHT ITEM ON THE £100 SHOPPING BUDGET — a bespoke-pupil lens is a prescription-adjacent purchase from an optical supplier, needs lead time, and needs a wear-time and handling protocol for the performer. Budget and order it separately, and early.

From the specification: FINE scales on exposed skin, and a separate discipline from the armour. The reptilian pupil is a CONTACT LENS on a human performer — the eye keeps its human shape, lids, lashes and expression, and only the pupil is wrong. It is the single highest-value element in her make-up. Locked against scale_portrait.png, not against the costume reference.